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Miyamoto

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(54) **METALWORKING PORTABLE CUTTING MACHINE**

(71) Applicant: **MAKITA CORPORATION**, Anjo (JP)

(72) Inventor: **Gen Miyamoto**, Anjo (JP)

(73) Assignee: **MAKITA CORPORATION**, Anjo (JP)

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B23D 59/00 (2006.01)

(52) **U.S. Cl.**
CPC **B23D 59/006** (2013.01)

(58) **Field of Classification Search**
CPC B23D 59/006
See application file for complete search history.

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Primary Examiner — Sean M Michalski

(74) *Attorney, Agent, or Firm* — Oliff PLC

(57) **ABSTRACT**

A metalworking portable cutting machine has a lid for an outlet of a dust collector box with improved operability. A metalworking portable cutting machine includes a cutting machine body including a blade, and a dust collector box to collect a chip resulting from cutting. The dust collector box includes an outlet to discharge the chip, and a lid to open or close the outlet. The lid includes a heat dissipater on a part of a surface of the lid.

12 Claims, 15 Drawing Sheets

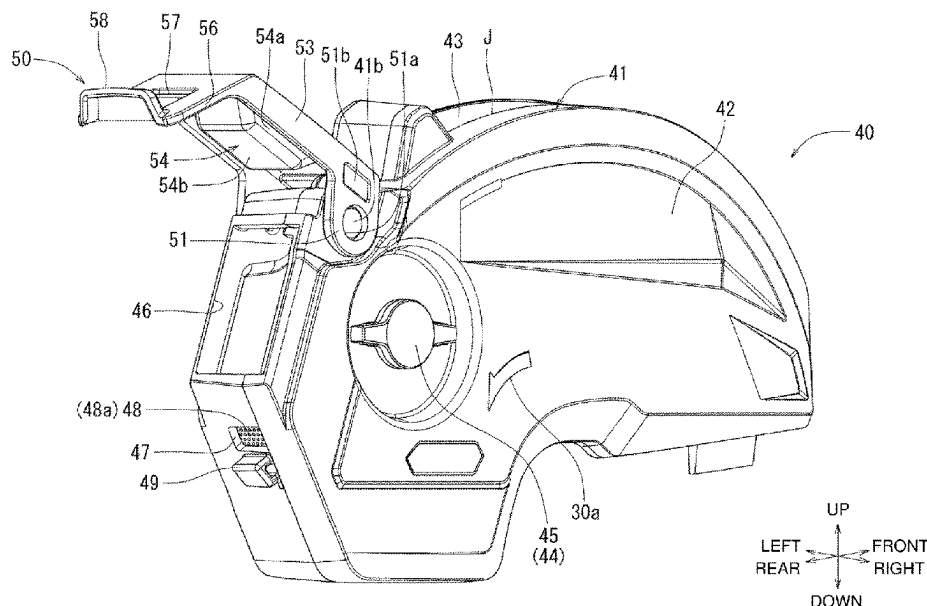


FIG. 1

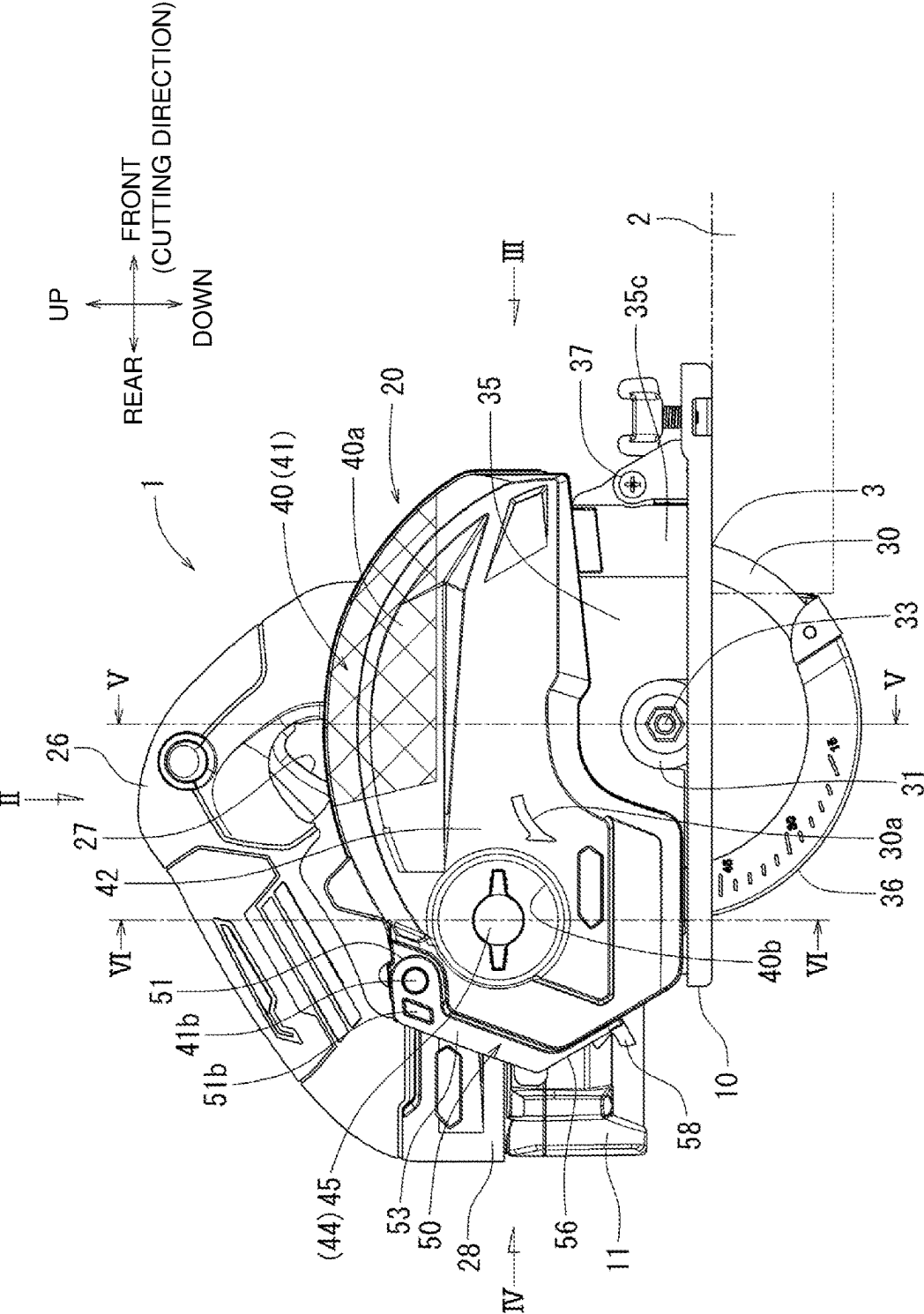


FIG. 2

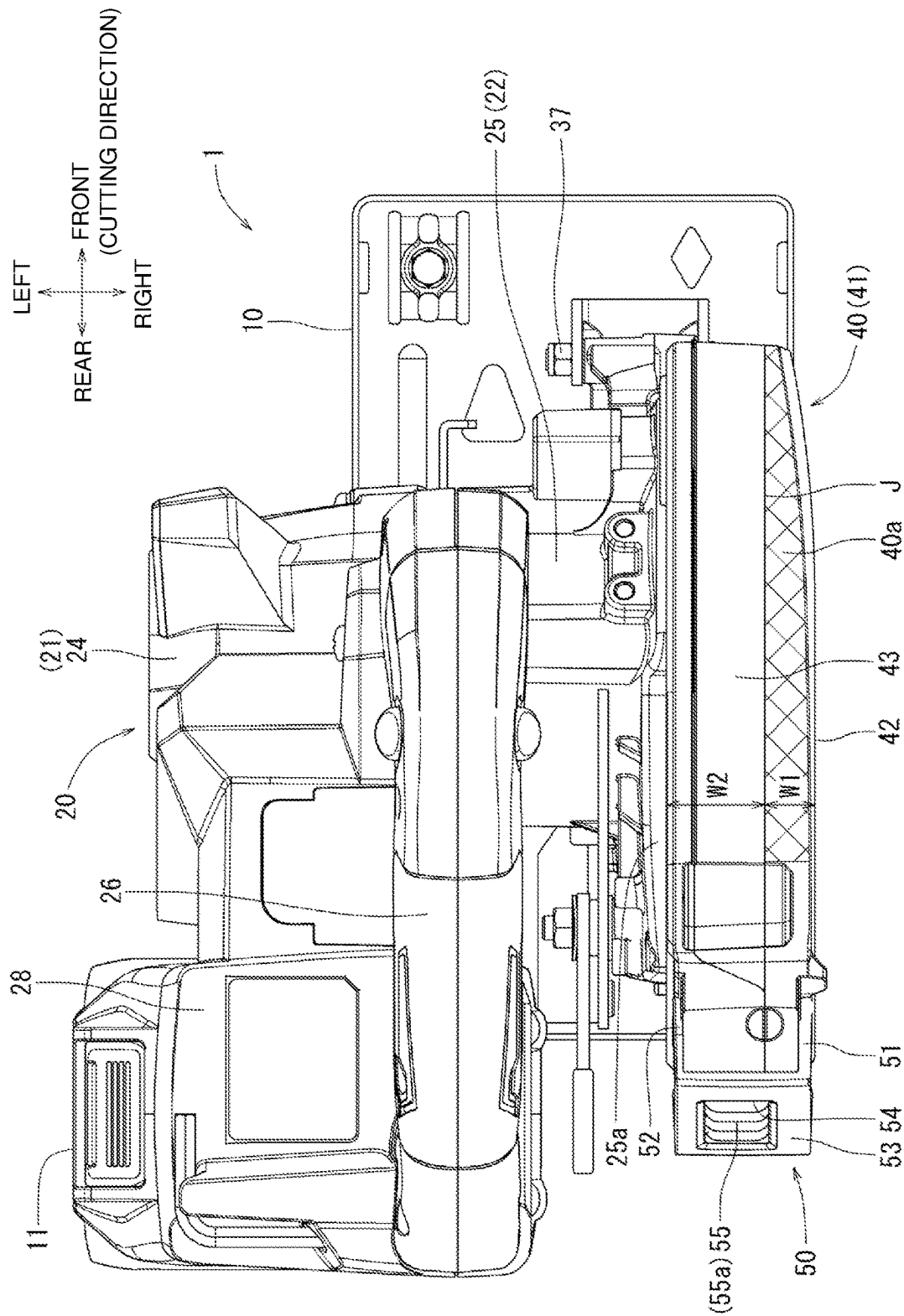


FIG. 3

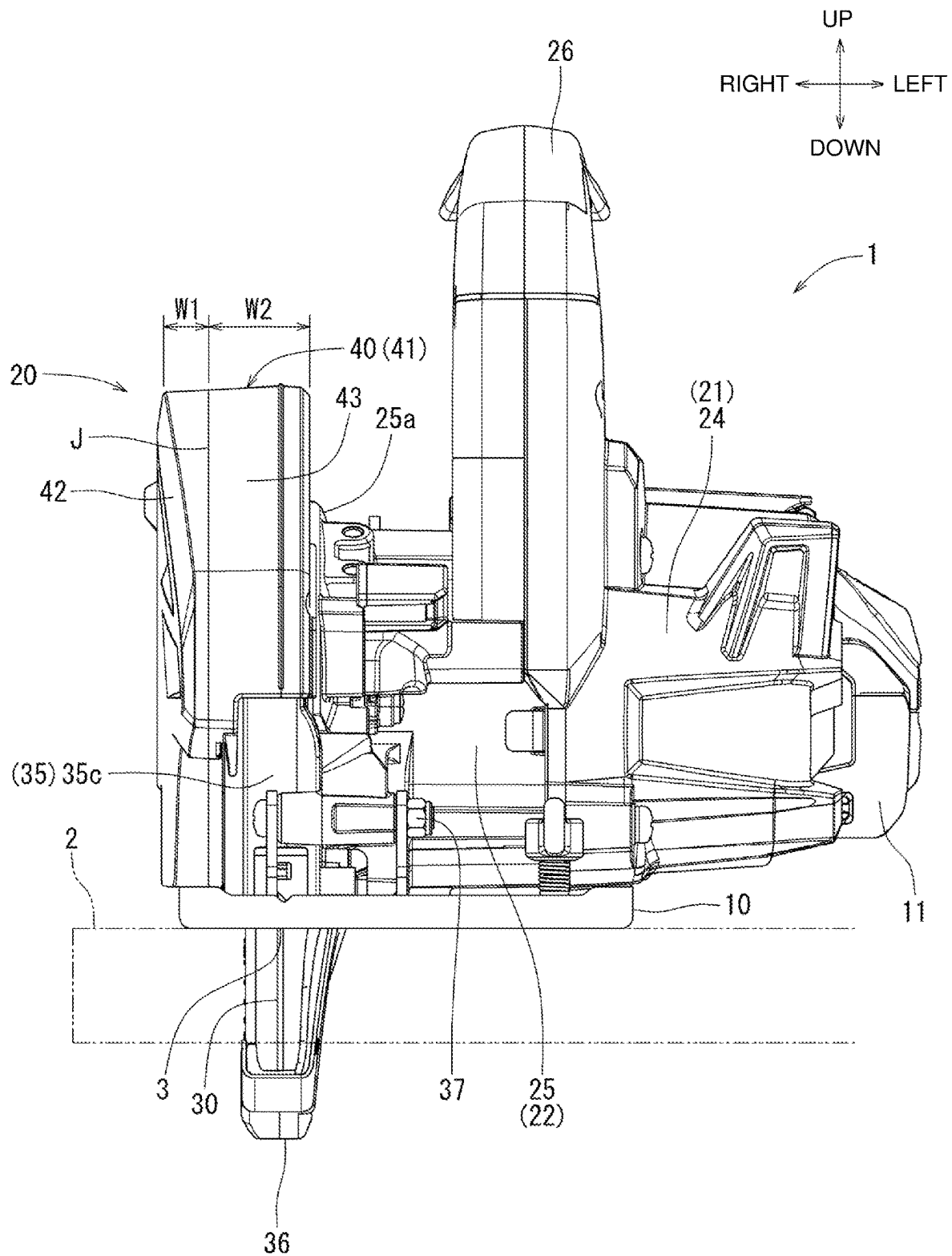


FIG. 4

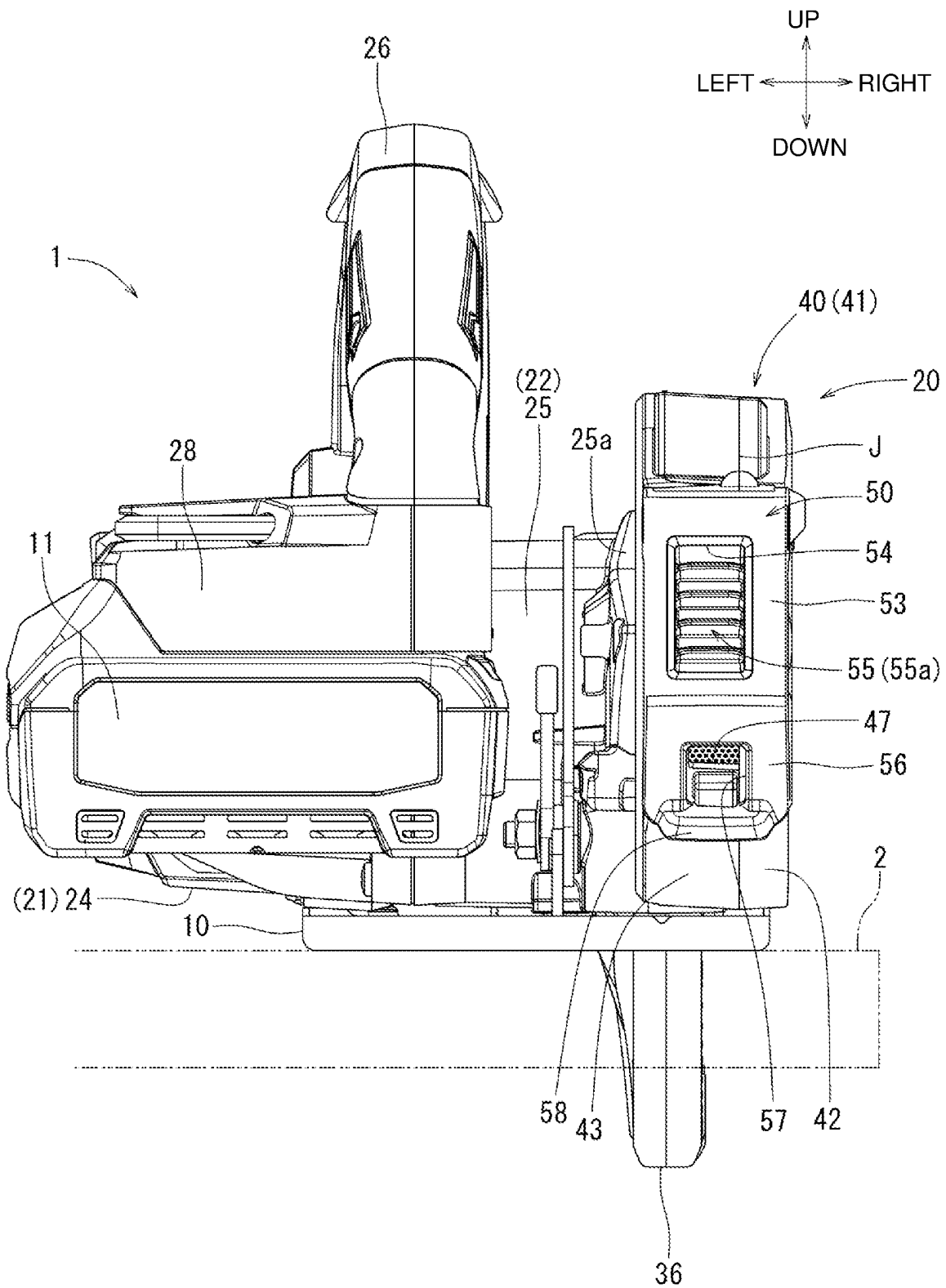


FIG. 5

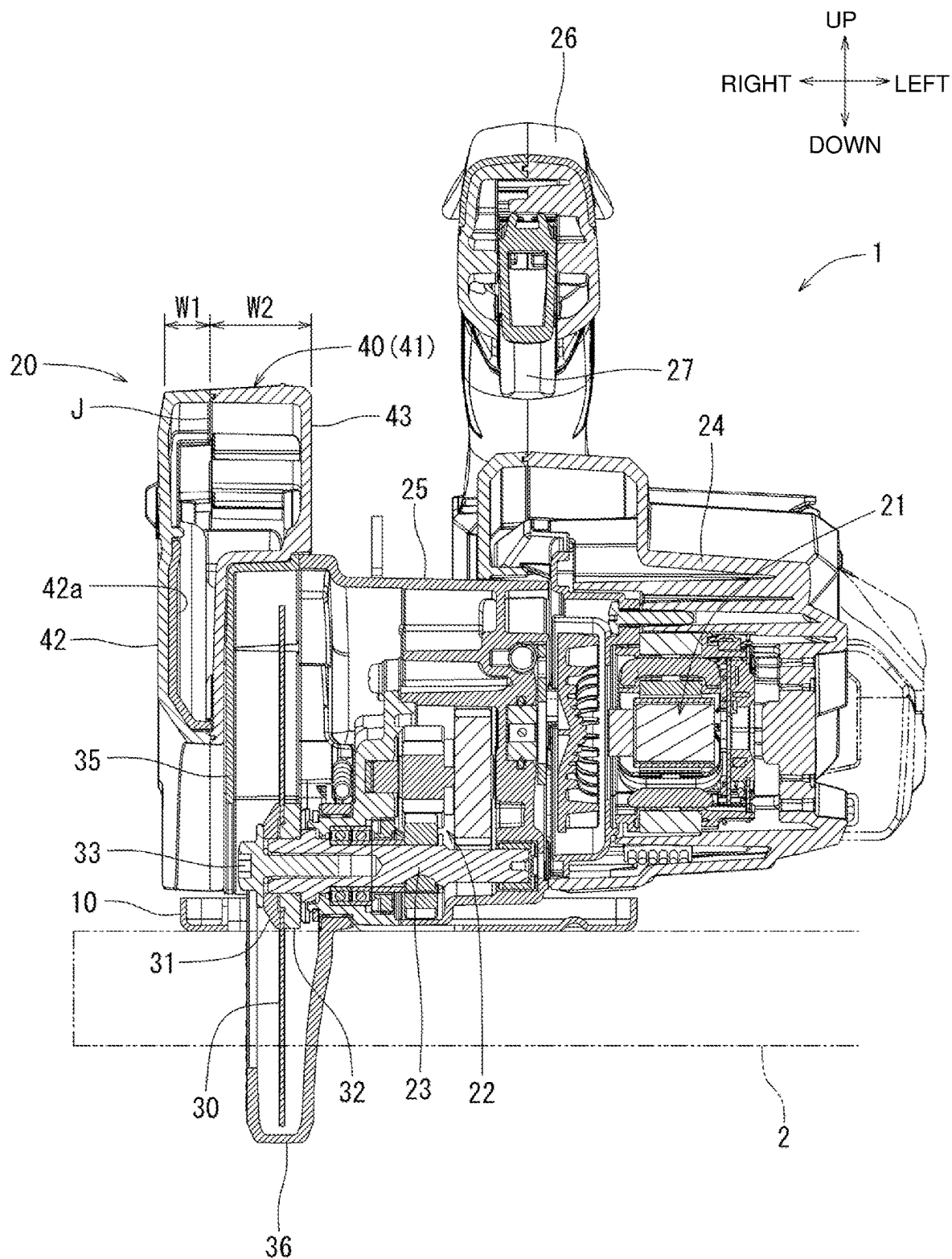


FIG. 6

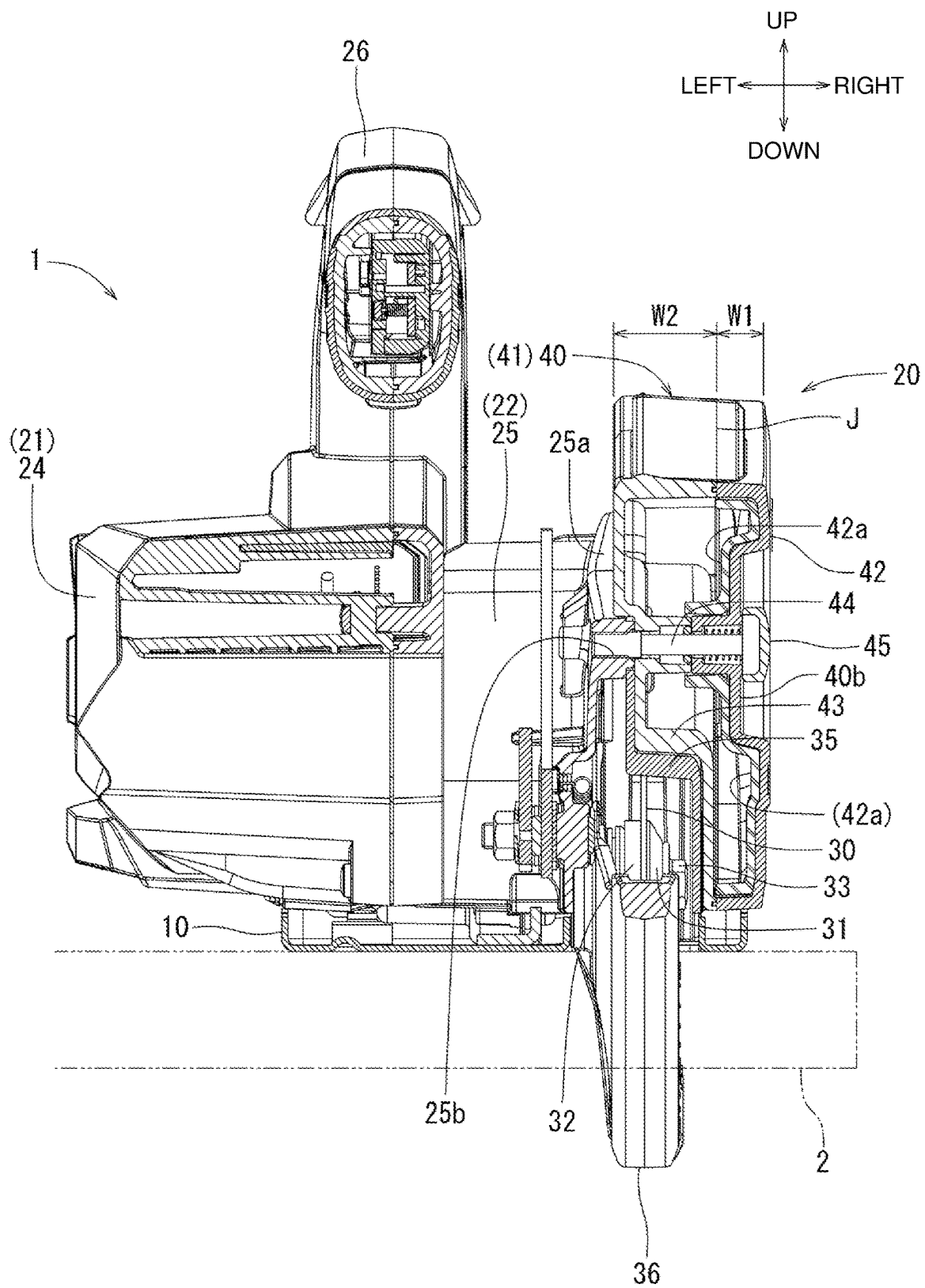
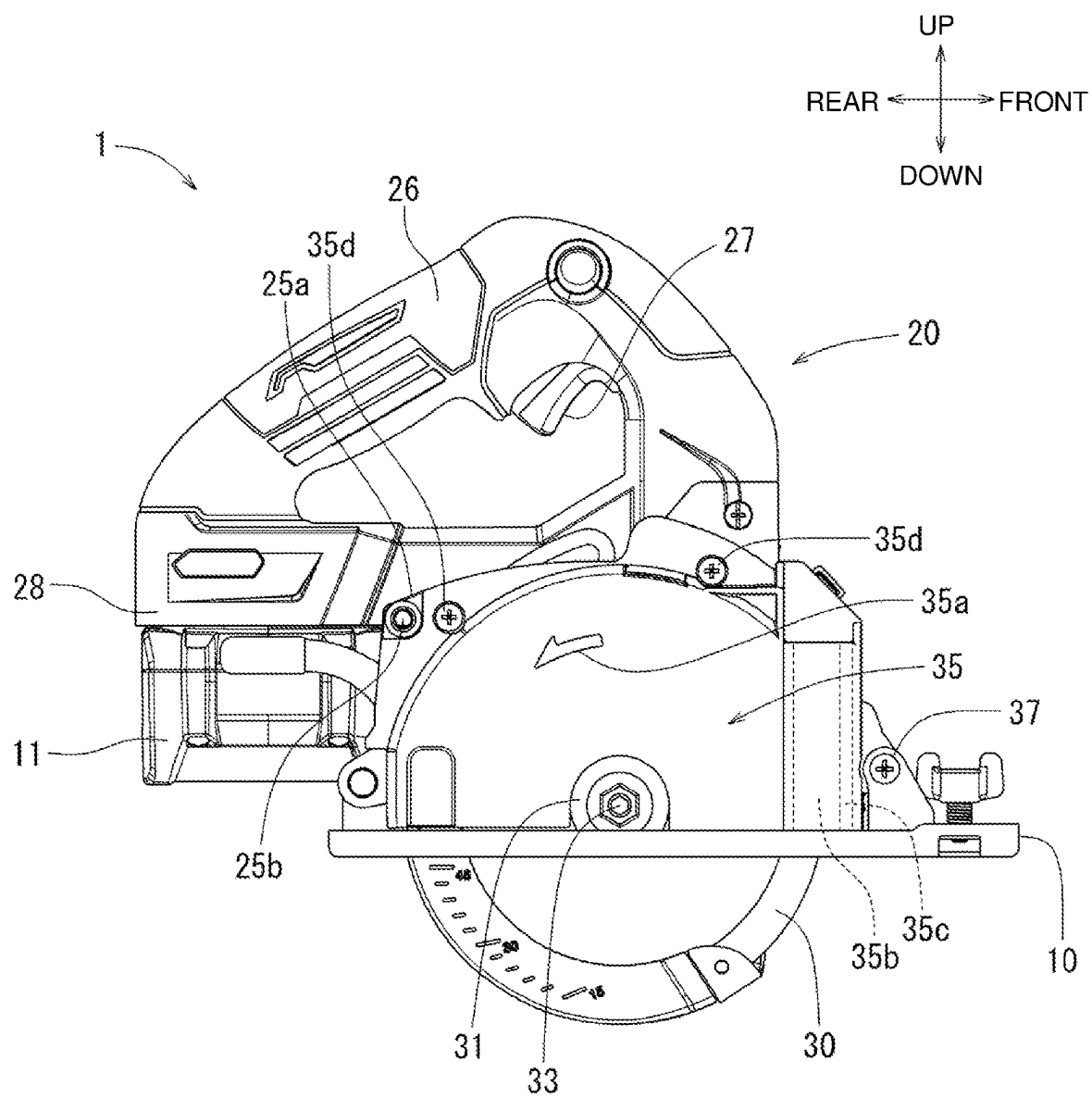


FIG. 7



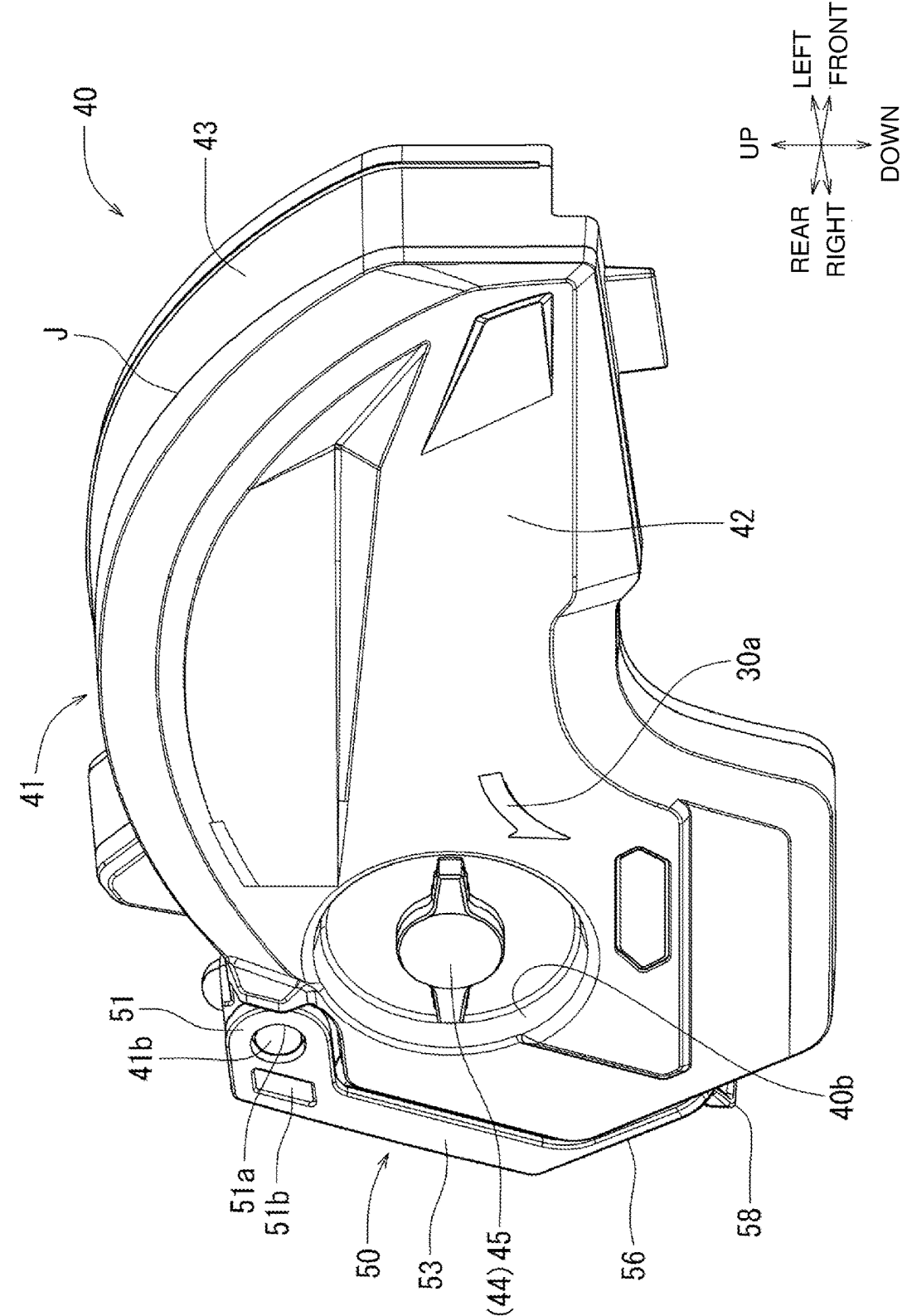


FIG. 8

FIG. 9

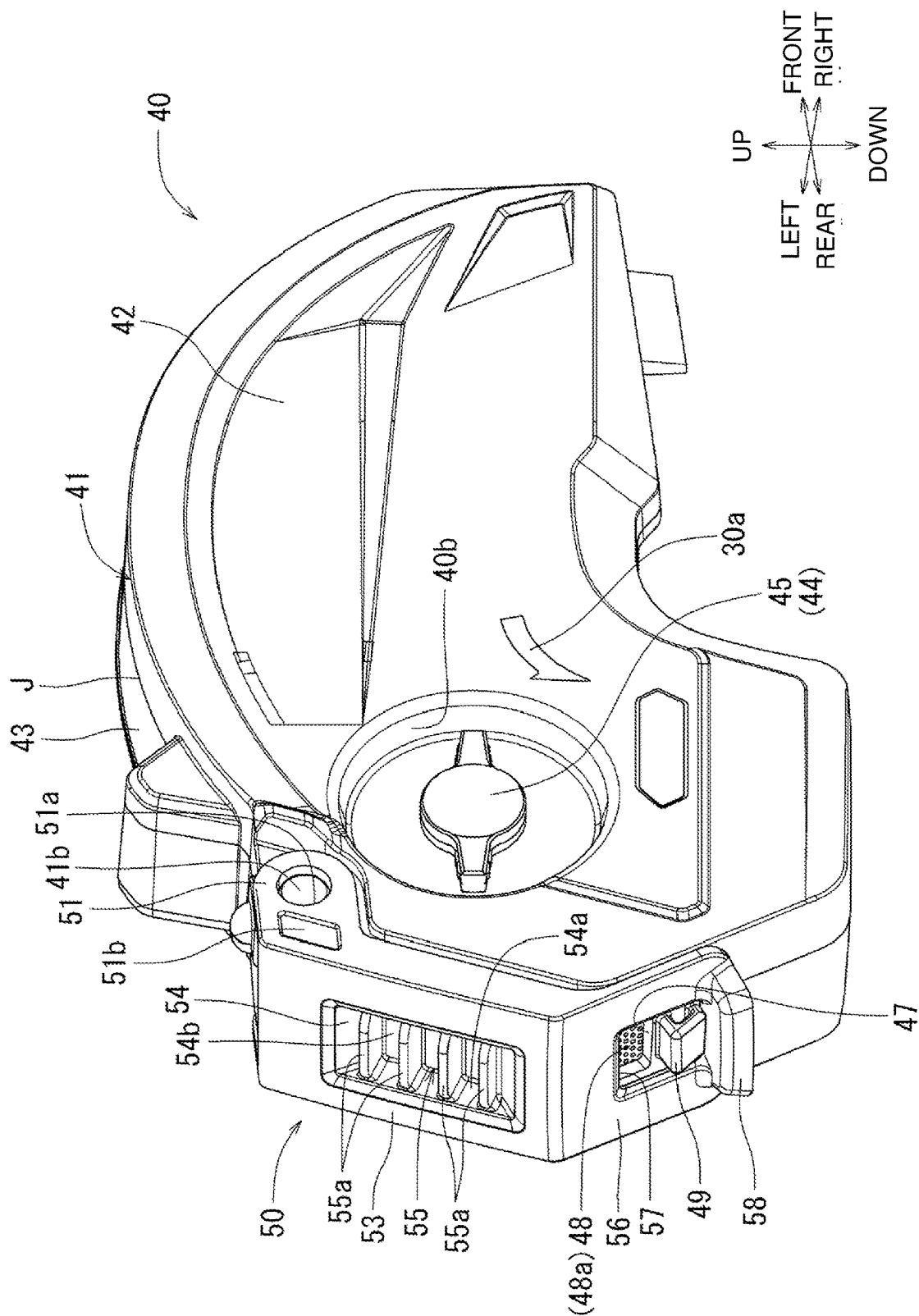
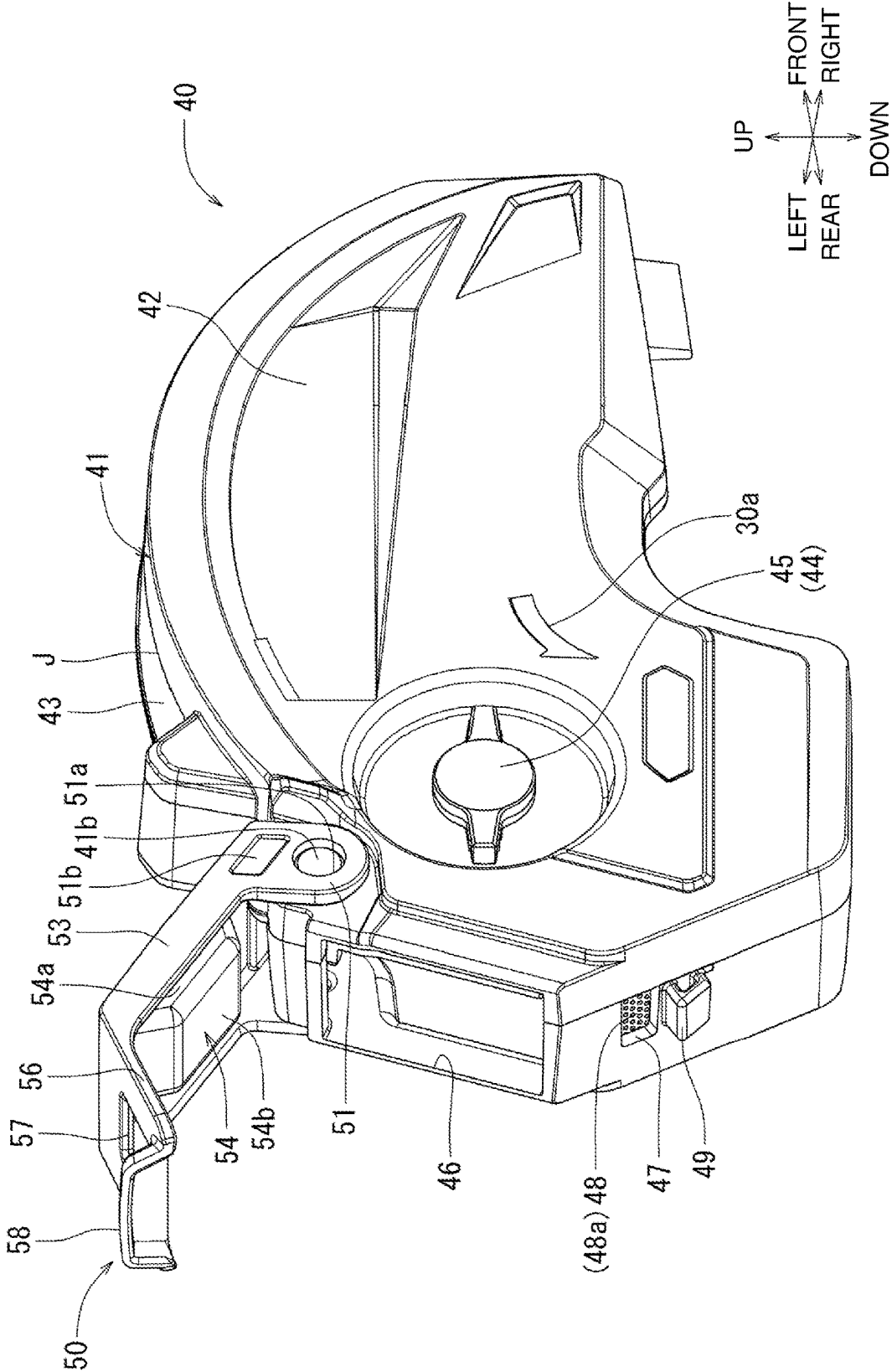
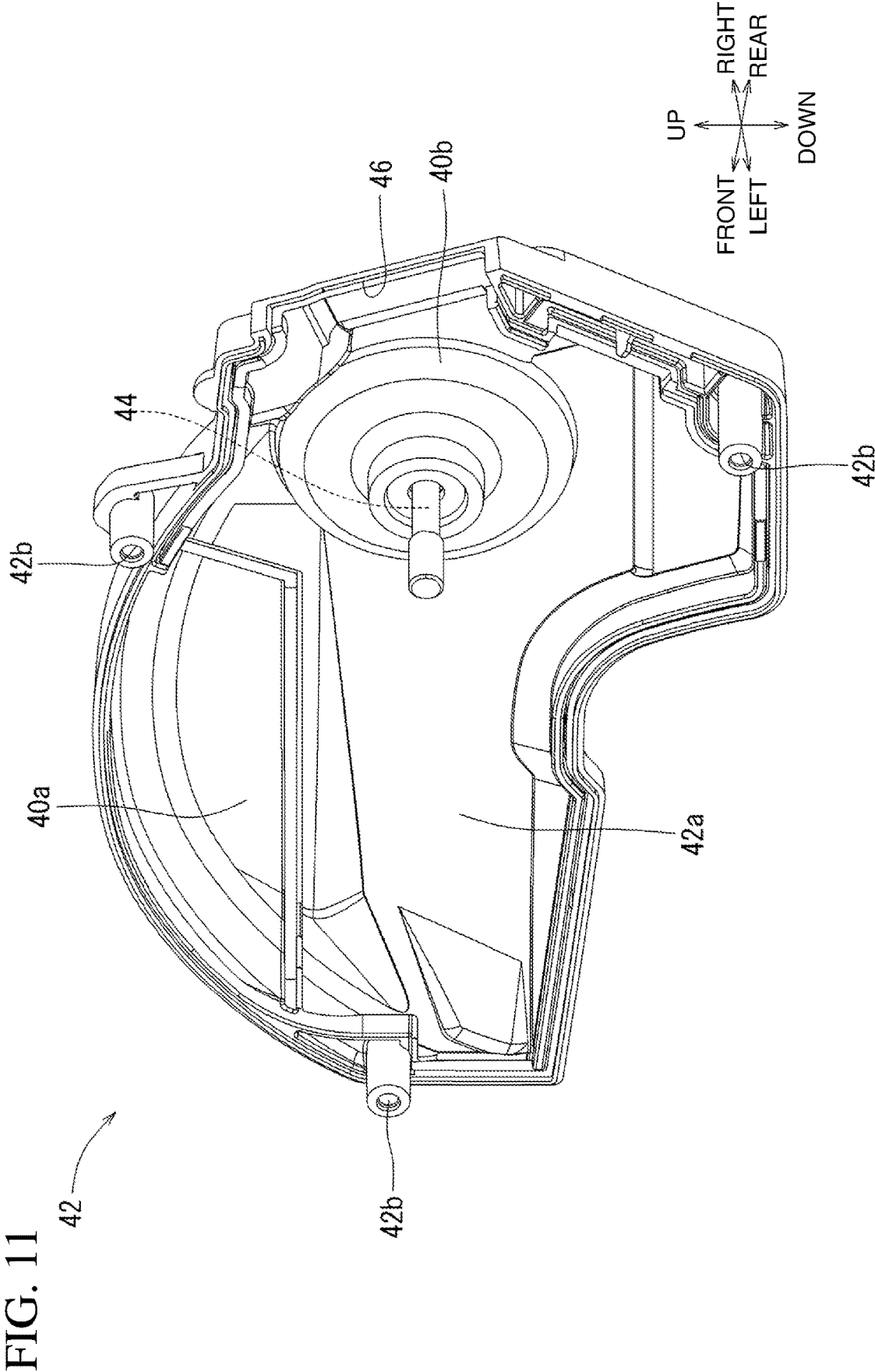


FIG. 10





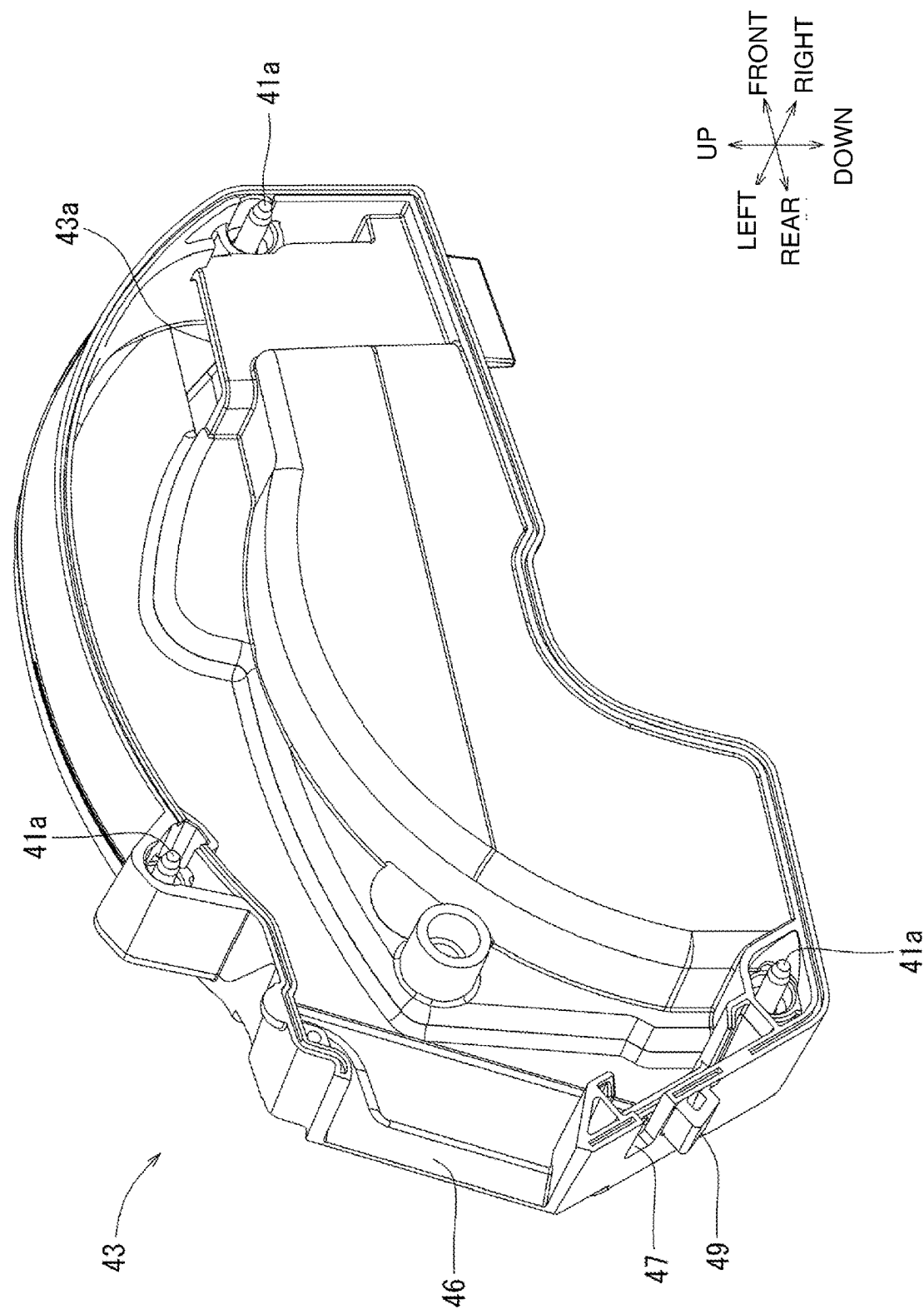


FIG. 12

FIG. 13

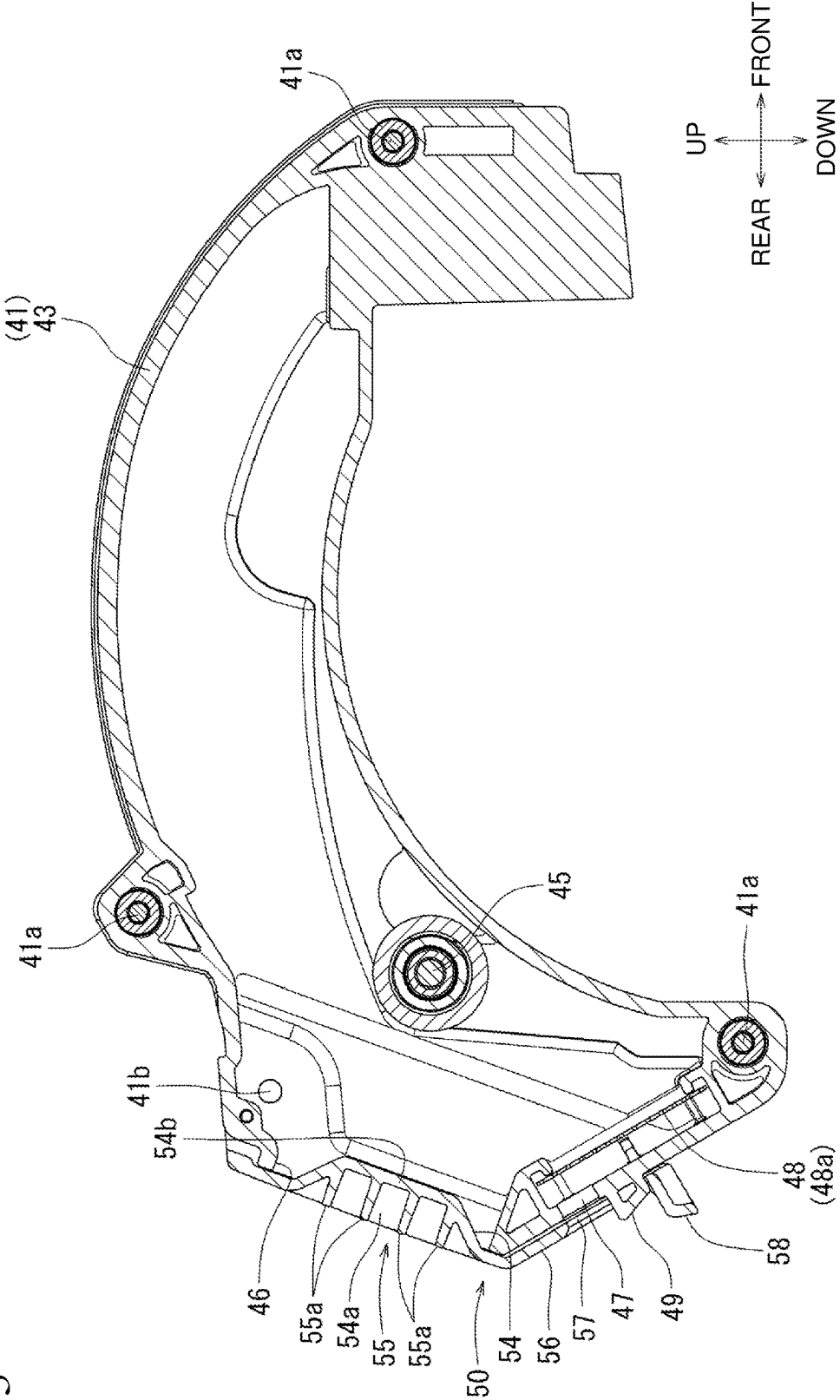


FIG. 14

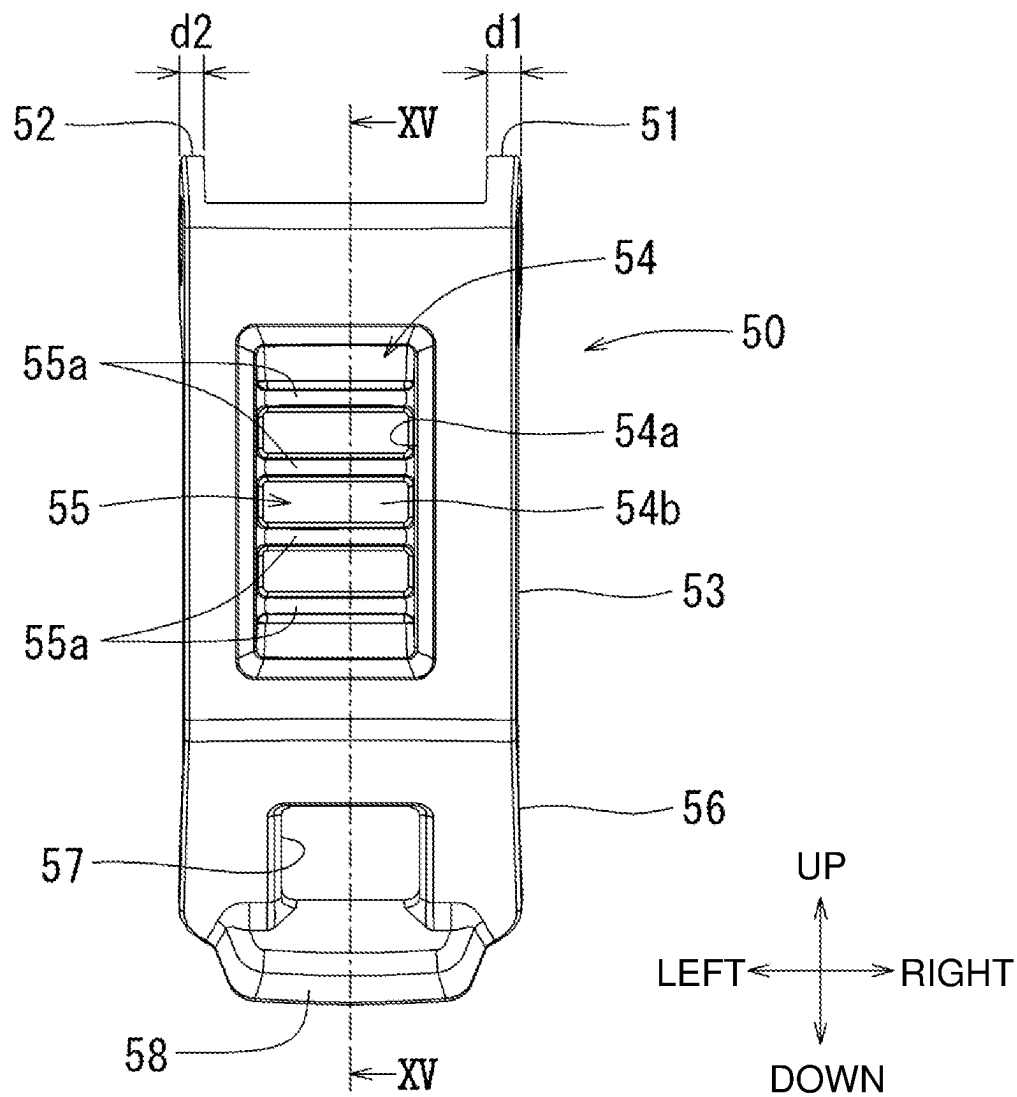
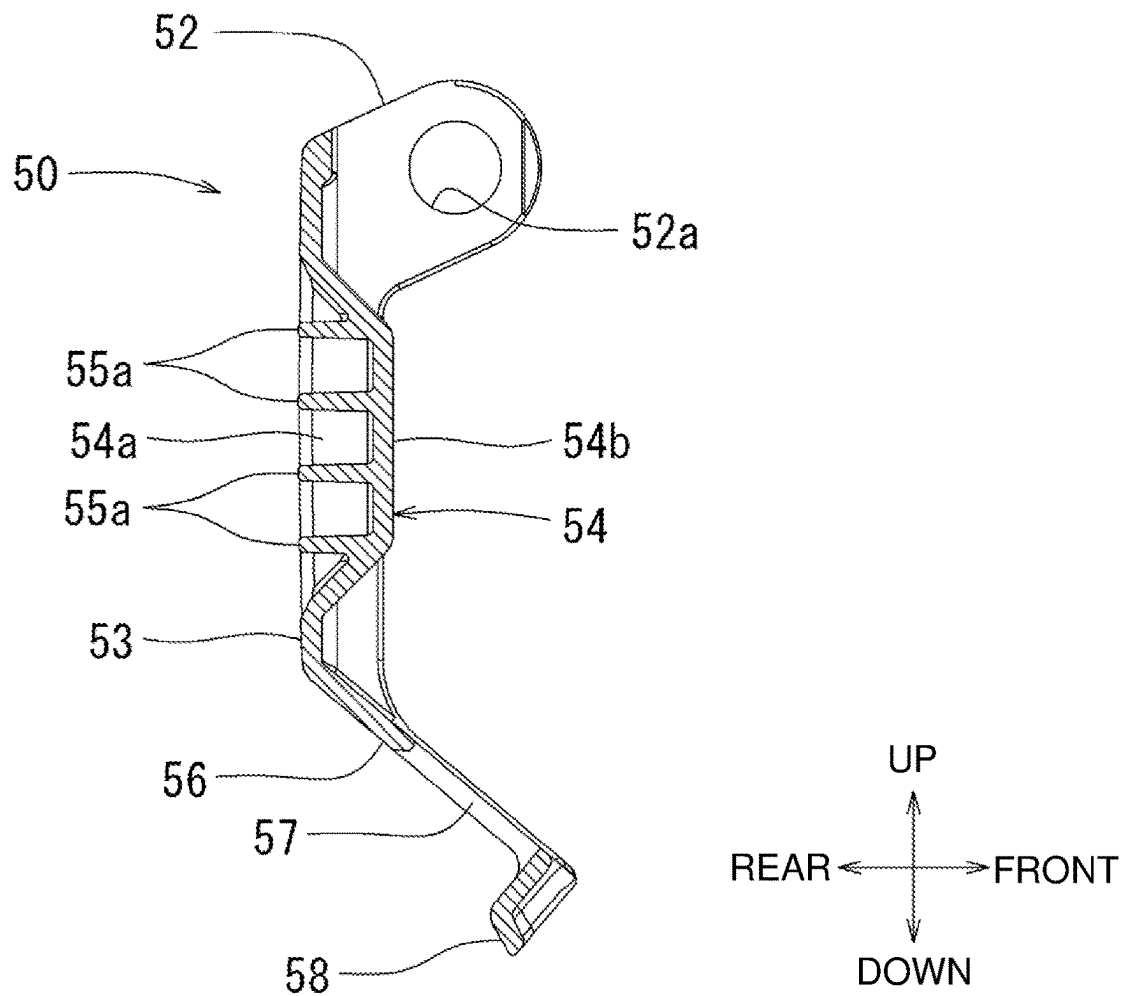


FIG. 15



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METALWORKING PORTABLE CUTTING MACHINE**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application claims the benefit of priority to Japanese Patent Application No. 2022-091564, filed on Jun. 6, 2022, the entire contents of which are hereby incorporated by reference.

BACKGROUND**1. Technical Field**

The present disclosure relates to a metalworking portable cutting machine to be held and operated by an operator with hands for cutting metal.

2. Description of the Background

For example, a metalworking portable cutting machine including a tipped saw blade for metal, referred to as a metal cutter, includes a dust collector box for collecting sparks resulting from cutting and high-temperature chips left after the sparks cool. As described in Japanese Unexamined Patent Application Publication No. 2019-198912, a dust collector box has a rear outlet through which collected chips are to be removed. The outlet can be open and closed with a lid. The user opens the lid to remove the chips in the dust collector box through the outlet. A tipped saw blade is a type of cutting blade having tips and a metal base prepared separately and then welded together.

BRIEF SUMMARY

The lid may be heated to high temperature with the chips in the dust collector box. The lid is thus to be open or closed with caution. The lid is thus to be more easily operable. One or more aspects of the present disclosure are directed to a metalworking portable cutting machine including a lid for an outlet of a dust collector box with improved operability.

A first aspect of the present invention provides a metalworking portable cutting machine, including:

- a cutting machine body including a blade; and
- a dust collector box to collect a chip resulting from cutting, the dust collector box including an outlet to discharge the chip, and a lid to open or close the outlet, the lid including a heat dissipater on a part of a surface of the lid.

In this structure, the heat dissipater allows efficient dissipation of heat from the lid. The lid is less likely to be hot and is easily operable when being open and closed.

A second aspect of the present invention provides a metalworking portable cutting machine, including:

- a cutting machine body including a blade; and
- a dust collector box to collect a chip resulting from cutting, the dust collector box including a box body, the box body including a first body comprising a first synthetic resin, the first body including a metal plate integral with an inner surface of the first body, and a second body comprising a second synthetic resin being more heat-resistant than the first synthetic resin.

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The use of metal in the first body alone reduces the weight of the dust collector box and allows the second body to improve the heat resistance of the dust collector box.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a right side view of a metalworking portable cutting machine according to an embodiment.

FIG. 2 is a plan view of the metalworking portable cutting machine as viewed in the direction indicated by arrow II in FIG. 1.

FIG. 3 is a front view of the metalworking portable cutting machine as viewed in the direction indicated by arrow III in FIG. 1.

FIG. 4 is a rear view of the metalworking portable cutting machine as viewed in the direction indicated by arrow IV in FIG. 1.

FIG. 5 is a cross-sectional view taken along line V-V in FIG. 1, as viewed in the direction indicated by arrows.

FIG. 6 is a cross-sectional view taken along line VI-VI in FIG. 1, as viewed in the direction indicated by arrows.

FIG. 7 is a right side view of the metalworking portable cutting machine with its dust collector box detached.

FIG. 8 is a perspective view of the dust collector box as viewed diagonally from the right front.

FIG. 9 is a perspective view of the dust collector box as viewed diagonally from the right rear.

FIG. 10 is a perspective view of the dust collector box as viewed diagonally from the right rear, with its lid open.

FIG. 11 is a perspective view of a first body as viewed diagonally from the left rear, showing the inside of a box body.

FIG. 12 is a perspective view of a second body as viewed diagonally from the right rear, showing the inside of the box body.

FIG. 13 is a longitudinal sectional view of the dust collector box, with the lid closed.

FIG. 14 is a rear view of the lid.

FIG. 15 is a longitudinal sectional view of the lid taken along line XV-XV in FIG. 14, as viewed in the direction indicated by arrows.

DETAILED DESCRIPTION

As shown in FIGS. 1 to 4, a metalworking portable cutting machine 1 according to an embodiment includes a tipped saw blade for metal and is also referred to as a metal cutter. The metalworking portable cutting machine 1 includes a base 10 and a cutting machine body 20. The base 10 is placed in contact with the upper surface of a workpiece 2. The cutting machine body 20 is supported on the upper surface of the base 10. The base 10 is a rectangular flat plate with its lower surface in contact with the upper surface of the workpiece 2. The user is behind the metalworking portable cutting machine 1 and moves the metalworking portable cutting machine 1 forward for cutting. The front herein refers to the cutting direction. The rear herein refers to the direction toward the user. The lateral direction is defined as viewed from the user.

As shown in FIGS. 5 to 7, the cutting machine body 20 includes an electric motor 21 as a driving source and a blade 30. The blade 30 is a circular plate rotatable by the electric motor 21. The electric motor 21 is, for example, a high-power brushless motor that is compact in a direction along its axis. Rotation from the electric motor 21 is output to an output shaft 23 through a reduction gear 22. A gear housing 25 is connected to the right end of a motor housing 24.

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A loop handle 26 is located above the joint between the motor housing 24 and the gear housing 25. A switch lever 27 is located on the inner periphery of the handle 26. The user turns on the switch lever 27 with the fingers holding the handle 26 to activate the electric motor 21.

A battery mount 28 is located at the rear of the handle 26. The battery mount 28 receives a battery pack 11. The battery pack 11 is, for example, a lithium-ion battery attachable in a slidable manner. The electric motor 21 is powered by the battery pack 11 mounted on the battery mount 28. The battery pack 11 detached from the battery mount 28 may be charged with a separate charger to allow repeated use.

As shown in FIGS. 5 to 7, a stationary cover 35 formed from a transparent resin is connected to a flange 25a at the right end of the gear housing 25. The stationary cover 35 is formed from polycarbonate and is connected to the gear housing 25 with two fastener screws 35d. The stationary cover 35 and the flange 25a on the gear housing 25 are connected together into a cover. The output shaft 23 protrudes into the stationary cover 35 to receive the blade 30. The blade 30 is held between an outer flange 31 and an inner flange 32 in a direction along its thickness. In this state, the blade 30 is fastened to the output shaft 23 with a fastener screw 33. The blade 30 is, for example, a circular saw blade for cutting metal materials, which is also referred to as a tipped saw blade.

The blade 30 has its substantially upper half (portion above the upper surface of the base 10) covered with the stationary cover 35 and the flange 25a on the gear housing 25. The blade 30 cutting into the workpiece 2 produces chips in a cutting portion 3. The chips are blown up by the rotating blade 30. The stationary cover 35 includes a rectangular dust collector duct 35b at its front. The dust collector duct 35b is located above the cutting portion 3. A metal plate 35c, which is a plate of steel, is located on the front inner peripheral surface of the dust collector duct 35b. The metal plate 35c protects the stationary cover 35 from high-temperature chips hitting the stationary cover 35. The chips blown up from the cutting portion 3 are thus guided through the dust collector duct 35b and flow into a dust collector box 40.

An arrow 35a on the right side of the stationary cover 35 indicates the rotation direction of the blade 30.

The blade 30 has its substantially lower half (portion below the lower surface of the base 10) covered with a movable cover 36. The movable cover 36 is rotatably supported on the right side of the gear housing 25. As the movable cover 36 rotates, the cutting edge of the blade 30 is exposed below the lower surface of the base 10. The movable cover 36 is urged in the closing direction. As the metalworking portable cutting machine 1 is moved for cutting, the movable cover 36 pressed against the workpiece 2 gradually opens against the urging force.

The cutting machine body 20 is supported on the upper surface of the base 10 with a swing support 37. The swing support 37 allows the cutting machine body 20 to swing vertically. The depth of a cut created by the blade 30 into the workpiece 2 is adjustable by changing the vertical swing position of the cutting machine body 20.

The dust collector box 40 is detachably attached to the right side of the stationary cover 35. The dust collector box 40 collects chips flowing into it from inside the stationary cover 35. FIGS. 8 to 10 show the dust collector box 40 detached from the right side of the stationary cover 35. The dust collector box 40 includes a box body 41 formed from a synthetic resin. An arrow 30a on the right side of the box body 41 indicates the rotation direction of the blade 30 (counterclockwise in FIG. 1).

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The box body 41 extends in the front-rear and vertical directions and is short in the lateral width direction (along the thickness of the blade 30). The box body 41 includes right and left halves. More specifically, the box body 41 includes a first body 42 on the right and a second body 43 on the left. The first body 42 and the second body 43 face each other and are connected together into the box body 41. The joint between the first body 42 and the second body 43 is denoted by character J in the drawings. As shown in FIG. 13, the first body 42 and the second body 43 are connected together with three fastener screws 41a in their front, rear, and upper portions. The three fastener screws 41a are placed through the second body 43 and tightened. The fastener screws 41a are thus invisible on the first body 42.

FIG. 11 shows the inner surface of the first body 42 separate from the second body 43. The first body 42 has three threaded holes 42b at its periphery for the fastener screws 41a to be tightened into. The first body 42 is formed from polycarbonate (first synthetic resin), which is a light-transmissive material. An aluminum heat-resistant plate 42a is connected to the inner surface of the first body 42 by press fitting. As shown in FIGS. 5 and 11, the heat-resistant plate 42a covers an area of the inner surface of the first body 42 extending rearward from the center of the blade 30. An area of the inner surface of the first body 42 extending frontward from the center of the blade 30 and a part of the intermediate area are covered with the heat-resistant plate 42a substantially in their lower half. The substantially upper half of these areas uncovered with the heat-resistant plate 42a and thus light-transmissive allows visual check of the volume of collected chips. In FIGS. 1 and 2, this light-transmissive area of the first body 42 (light-transmissive portion 40a) for the visual check of the volume of chips inside is cross-hatched.

As shown in FIGS. 6 and 11, the inner surface of the first body 42 is covered with the heat-resistant plate 42a throughout its vertical range in its intermediate area and an area rearward from the center of the blade 30. This improves the heat resistance of the first body 42 throughout its vertical range.

FIG. 12 shows the inner surface of the second body 43 separate from the first body 42. The second body 43 is formed from glass fiber-containing polyamide (second synthetic resin), which is a highly heat-resistant material. As shown in FIGS. 2, 3, 5, and 6, the second body 43 has a width W2 greater than a width W1 of the first body 42 in the direction along the thickness of the blade 30. The greater width W2 of the highly heat-resistant second body 43 further improves the heat resistance of the dust collector box 40.

As shown in FIG. 12, the second body 43 has a dust collecting port 43a at the front on its inner surface. The chips blown up into the stationary cover 35 are carried with an airflow generated by the rotating blade 30, guided through the dust collector duct 35b, and collected into the box body 41 through the dust collecting port 43a. The dust collecting port 43a corresponds to the inlet of the dust collector box 40. The dust collecting port 43a is located above the dust collector duct 35b.

As shown in FIG. 6, the dust collector box 40 is fastened to the stationary cover 35 with a fastener screw 44. As shown in FIGS. 6 and 7, the fastener screw 44 is tightened into a threaded hole 25b in the flange 25a on the right side of the gear housing 25. As shown in FIGS. 1 and 8 to 10, the fastener screw 44 is fitted with a large knob 45 at its right end. This improves the operability of the fastener screw 44 in tightening and loosening operations. The fastener screw 44 is loosened and removed from the threaded hole 25b to detach the dust collector box 40. The knob 45 is received in

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a circular recess **40b** on the right side of the box body **41**. This structure allows the dust collector box **40** to be compact in the lateral width direction without the knob **45** protruding.

As shown in FIGS. **10** and **13**, the box body **41** has an outlet **46** for discharging chips and a vent **47** at its rear. The outlet **46** is a rectangular opening. The outlet **46** can be open and closed with a lid **50**. As shown in FIGS. **8** to **10**, the lid **50** is supported in a manner vertically pivotable about a support shaft **41b** located at the rear of the box body **41**. The support shaft **41b** protrudes both to the right and to the left.

Similarly to the second body **43** in the box body **41**, the lid **50** is formed from glass fiber-containing polyamide. As shown in FIGS. **14** and **15**, the lid **50** has a pair of hinges, or a right hinge **51** and a left hinge **52**, and an arm **53**. The pair of right and left hinges **51** and **52** are connected to right and left portions of the support shaft **41b**. The arm **53** extends from ends of the hinges **51** and **52** at an angle with the hinges **51** and **52**. The right and left hinges **51** and **52** have connecting holes **51a** and **52a**. The support shaft **41b** is placed through each of the connecting holes **51a** and **52a** to support the lid **50** in a vertically rotatable manner.

As shown in FIG. **14**, the right hinge **51** has a thickness **d1** greater than a thickness **d2** of the left hinge **52**. The right hinge **51** is thus harder than the left hinge **52**. This facilitates attachment of the lid **50** to the box body **41**. The harder right hinge **51** is connected to the support shaft **41b** first. The less hard left hinge **52** can thus be easily connected to the support shaft **41b** later. As shown in FIGS. **8** to **10**, the right hinge **51** has a recess **51b** on its right side. This reduces the cost for molding.

As shown in FIGS. **13** to **15**, the arm **53** has a recess **54**. The recess **54** is recessed toward the hinges **51** and **52** (frontward). The recess **54** includes a wall **54a** and a bottom **54b**. The wall **54a** is looped and protrudes frontward. The wall **54a** has its end covered with the bottom **54b**. The recess **54** receives a heat dissipater **55**. The heat dissipater **55** enhances the heat dissipation of the lid **50**. This improves the operability of the lid **50** in opening and closing operations.

The heat dissipater **55** includes multiple (four in the present embodiment) ridges **55a**. The four ridges **55a** are received in the recess **54**. Each ridge **55a** extends laterally. The four ridges **55a** are regularly spaced and extend parallel to one another. The four ridges **55a** extend from the bottom **54b** of the recess **54** perpendicularly to the direction of rotation of the lid **50**, or parallel to the tangent to an arc centered on the support shaft **41b**. The ends of the four ridges are connected to the wall **54a** of the recess **54**. The wall **54a** connects the ends of the four ridges **55a** together, improving the rigidity of each ridge **55a** as a support. The four ridges **55a** also improve the rigidity of the recess **54**.

The four ridges **55a** and the recess **54** increase the external surface area of the lid **50** to enhance the heat dissipation of the lid **50**. As shown in FIG. **13**, the recess **54** is located in the outlet **46** when the lid **50** is closed. The dust collector box **40** can thus have a compact rear portion when the lid **50** is closed. The recess **54** and the heat dissipater **55** located in the outlet **46** also facilitate the discharge of heat from the outlet **46** and its surroundings.

The lid **50** includes an engagement tab **56**. The engagement tab **56** extends from an end of the arm **53** at an angle toward the hinges **51** and **52**. The engagement tab **56** has a rectangular engagement hole **57**. The engagement hole **57** has its lower edge elastically engaged with an engagement portion **49** on the rear surface of the dust collector box **40** when the lid **50** is closed. This elastic engagement between the lower edge of the engagement hole **57** and the engagement portion **49** holds the lid **50** being closed at the position.

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The engagement portion **49** has its rear surface sloping frontward to the lower end. This allows easier operation in opening and closing the lid **50** (the operability of the lid **50** in opening and closing operations) for causing the lower edge of the engagement hole **57** to be engaged with or disengaged from the engagement portion **49**.

As shown in FIGS. **9**, **10**, and **13**, the dust collector box **40** has the vent **47** in its rear surface. The vent **47** is covered with a metal air-vent plate **48** commonly referred to as perforated metal. The air-vent plate **48** has multiple small air vents **48a**. The air-vent plate **48** improves the heat resistance of the vent **47** and an airflow through the vent **47**. As shown in FIGS. **9** and **13**, the vent **47** faces an upper portion of the engagement hole **57** when the lid **50** is closed. The vent **47** is not blocked when the lid **50** is closed. This allows smooth airflow in the box body **41** and efficient discharge of heat from the box body **41** through the vent **47** when the lid **50** is closed.

A projection **58** is located at the lower end of the engagement tab **56**. The projection **58** extends away from the box body **41**. The projection **58** is pulled away from the box body **41** to detach the engagement portion **49** from the lower edge of the engagement hole **57**. The projection **58** is then pulled upward to open the lid **50**. To close the lid **50**, the projection **58** is pulled downward toward the box body **41**. This engages the lower edge of the engagement hole **57** with the engagement portion **49**. The projection **58** improves the operability of the lid in opening and closing operations.

The metalworking portable cutting machine **1** according to the present embodiment includes the dust collector box **40** with the outlet **46** and the lid **50** to open or close the outlet **46**. The lid **50** includes the heat dissipater **55**. This allows efficient dissipation of heat from the lid through the heat dissipater **55** and thus improves the operability of the lid **50** in opening and closing operations.

The dust collector box **40** in the embodiment includes the box body **41** having right and left halves, or the first body **42** and the second body **43**. The first body **42** is formed from polycarbonate (first synthetic resin) with the aluminum heat-resistant plate **42a** integral with its inner surface. The second body **43** is formed from glass fiber-containing polyamide (second synthetic resin), which is a material more heat-resistant than polycarbonate. The use of the metal component in the first body **42** alone reduces the weight of the dust collector box **40** and allows the second body **43** to improve the heat resistance of the dust collector box **40**.

The heat dissipater **55** in the embodiment includes the multiple ridges **55a**. This increases the surface area of the lid **50** to allow efficient dissipation of heat from the lid **50**.

The multiple ridges **55a** in the embodiment are parallel to one another. This creates the channels for outside air to flow between the ridges **55a** to enhance the heat dissipation of the lid **50**.

The lid **50** in the embodiment is rotatably connected to the box body **41** of the dust collector box **40**. The multiple ridges **55a** extend perpendicularly to the direction of rotation of the lid **50**. The lid **50** includes the wall **54a** connecting the ends of the multiple ridges **55a** together. This improves the rigidity of the multiple ridges **55a** as a support.

The lid **50** in the embodiment includes the hinges **51** and **52** rotatably connected to the box body **41** of the dust collector box **40**. The arm **53** extends from the ends of the hinges **51** and **52** at an angle with the hinges **51** and **52**. The arm **53** has the recess **54** recessed toward the hinges **51** and **52**. The arm **53** with this structure is less likely to deform and thus improves the rigidity of the lid **50** without increas-

ing its size. The arm **53** having the recess **54** with its increased surface area also enhances heat dissipation.

The recess **54** in the embodiment is located in the outlet **46** when the lid **50** is closed. The dust collector box **40** can have a compact rear portion when the lid **50** is closed.

The recess **54** in the embodiment includes the wall **54a** that is looped and extends from the arm **53** toward the hinges **51** and **52** and the bottom **54b** covering the end of the wall **54a**. The heat dissipater **55** includes the multiple ridges **55a** protruding from the bottom **54b** toward the arm **53**. The heat dissipater **55** includes the multiple ridges **55a** received in the recess **54** and thus is compact.

The lid **50** in the embodiment includes the engagement tab **56** extending from the end of the arm **53** at an angle toward the hinges **51** and **52**. The engagement tab **56** has the engagement hole **57** engageable with the engagement portion **49** of the dust collector box **40**. This structure holds the lid **50** being closed at the position. The hinges **51** and **52**, the arm **53**, and the engagement tab **56** are arranged in a curve along the rear surface of the box body **41** to improve the rigidity of the lid **50**. This structure improves the operability of the lid **50** in opening and closing operations.

The dust collector box **40** in the embodiment has the vent **47** to allow entry of outside air into the dust collector box **40**. The vent **47** is covered with the metal air-vent plate **48** with multiple air vents **48a**. The engagement hole **57** in the lid **50** faces the vent **47** (air vents **48a**). This improves airflow through the vent **47** (air vents **48a**) when the lid **50** is closed.

The lid **50** in the embodiment includes the projection **58** extending from the end of the engagement tab **56** away from the box body **41**. This structure improves the operability of the lid **50** in opening and closing operations.

The polycarbonate (first synthetic resin) in the first body **42** in the embodiment is light-transmissive. This allows visual check of the volume of collected chips through the light-transmissive portion **40a**.

The box body **41** in the embodiment includes the first body **42** in a first direction along the thickness of the blade **30** and the second body **43** in a second direction along the thickness of the blade **30**. The second body **43** has the width **W2** along the thickness greater than the width **W1** of the first body **42** along the thickness. This allows the dust collector box to have high heat resistance.

The second body **43** in the dust collector box **40** in the embodiment is formed from a material containing polyamide as the second synthetic resin and also containing glass fiber. This allows the second body **43** to have high heat resistance.

The embodiment described above may be modified variously. For example, although the four ridges **55a** on the heat dissipater **55** extend laterally in the above embodiment, the ridges **55a** may extend vertically. In some embodiments, the heat dissipater **55** may include multiple ridges arranged in a lattice pattern.

Although the multiple ridges **55a** are received in the recess **54** with their tops lower than the surface of the arm **53** in the above embodiment, the recess **54** may be eliminated, or the multiple ridges may protrude from a part of the surface of the arm **53**.

Although the thicknesses **d1** and **d2** of the right and left hinges **51** and **52** of the lid **50** are different in the above embodiment, the hinges **51** and **52** may have the same thickness. The recess **51b**, which is a lightening portion, may be eliminated.

Although the lid **50** vertically pivots to be open and closed in the above embodiment, the lid may be vertically slidable to be open and closed and may include the heat dissipater **55** in the above embodiment.

The second body **43** may be formed from a highly heat-resistant resin in place of glass fiber-containing polyamide.

Similarly to the second body **43**, the lid **50** may be formed from glass fiber-containing polyamide or any other highly heat-resistant resin.

The heat-resistant plate **42a** in the first body **42** may be a heat-resistant plate of iron rather than aluminum.

The first body **42** in the box body **41** may also be formed from glass fiber-containing polyamide, similarly to the second body **43**, except at a window for the visual check of the volume of collected chips. In this case, the heat-resistant plate **42a** may be eliminated. The window for the visual check of the volume of collected chips may be a separate transparent plate formed from a light-transmissive polycarbonate and fitted into the window. The box body may be a one-piece structure, for example produced by blow molding, rather than having right and left halves.

The metalworking portable cutting machine may include a diamond wheel or a grinding disc rather than being a metal cutter including a tipped saw blade for metal.

REFERENCE SIGNS LIST

- 1 metalworking portable cutting machine (metal cutter)
- 2 workpiece
- 3 cutting portion
- 10 base
- 11 battery pack
- 20 cutting machine body
- 21 electric motor (brushless motor)
- 22 reduction gear
- 23 output shaft
- 24 motor housing
- 25 gear housing
- 25a flange
- 25b threaded hole
- 26 handle
- 27 switch lever
- 28 battery mount
- 30 blade
- 30a arrow (rotation direction of blade 30)
- 31 outer flange
- 32 inner flange
- 33 fastener screw
- 35 stationary cover
- 35a arrow (rotation direction of blade 30)
- 35b dust collector duct
- 35c metal plate
- 35d fastener screw
- 36 movable cover
- 37 swing support
- 40 dust collector box
- 40a light-transmissive portion
- 40b recess
- 41 box body
- 41a fastener screw
- 41b support shaft
- 42 first body
- 42a heat-resistant plate
- 42b threaded hole
- W1 width of first body 42
- 43 second body

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43a dust collecting port
 W2 width of second body 43
 44 fastener screw
 45 knob
 46 outlet
 47 vent
 48 air-vent plate
 48a air vent
 49 engagement portion
 50 lid
 51 hinge (right)
 51a connecting hole
 51b recess (lightening portion)
 d1 thickness of hinge 51
 52 hinge (left)
 52a connecting hole
 d2 thickness of hinge 52
 53 arm
 54 recess
 54a wall
 54b bottom
 55 heat dissipater
 55a ridge
 56 engagement tab
 57 engagement hole
 58 projection
 What is claimed is:
 1. A metalworking portable cutting machine, comprising:
 a cutting machine body including a blade; and
 a dust collector box to collect a chip resulting from
 cutting, the dust collector box including a box body, the
 box body including
 a first body comprising a first synthetic resin, the first
 body including a metal plate integral with an inner
 surface of the first body, and
 a second body comprising a second synthetic resin
 being more heat-resistant than the first synthetic
 resin.
 2. The metalworking portable cutting machine according
 to claim 1, wherein
 the dust collector box includes
 an outlet to discharge the chip, and
 a lid to open or close the outlet, and
 the lid includes
 a hinge rotatably connected to the box body,
 an arm extending from an end of the hinge at an angle
 with the hinge, and
 a recess recessed from the arm toward the hinge.
 3. The metalworking portable cutting machine according
 to claim 2, wherein
 the recess is located in the outlet with the lid being closed.

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4. The metalworking portable cutting machine according
 to claim 2, wherein
 the recess includes
 a wall being looped and extending from the arm toward
 the hinge, and a bottom covering an end of the wall.
 5. The metalworking portable cutting machine according
 to claim 2, wherein
 the lid includes an engagement tab extending from an end
 of the arm at an angle toward the hinge, and
 the engagement tab has an engagement hole engageable
 with an engagement portion of the dust collector box.
 6. The metalworking portable cutting machine according
 to claim 5, wherein
 the dust collector box includes a metal air-vent plate
 having a plurality of air vents, and
 the engagement hole faces the metal air-vent plate.
 7. The metalworking portable cutting machine according
 to claim 5, wherein
 the lid includes a projection extending from an end of the
 engagement tab away from the box body.
 8. The metalworking portable cutting machine according
 to claim 1, wherein
 the first synthetic resin in the first body is light-transmis-
 sive.
 9. The metalworking portable cutting machine according
 to claim 8, wherein
 the first body is located in a first direction along a
 thickness of the blade,
 the second body is located in a second direction along the
 thickness of the blade, and
 the second body has a width along the thickness greater
 than a width of the first body along the thickness.
 10. The metalworking portable cutting machine according
 to claim 1, wherein
 the first body is located in a first direction along a
 thickness of the blade,
 the second body is located in a second direction along the
 thickness of the blade, and
 the second body has a width along the thickness greater
 than a width of the first body along the thickness.
 11. The metalworking portable cutting machine according
 to claim 1, wherein
 the second body comprises a material containing poly-
 amide as the second synthetic resin and containing
 glass fiber.
 12. The metalworking portable cutting machine according
 to claim 8, wherein
 the second body comprises a material containing poly-
 amide as the second synthetic resin and containing
 glass fiber.

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